

Education

New York University, New York, NY

Aug 2024 – May 2026

Master of Science in Computer Engineering (GPA: 3.9/4.0)

Coursework: System Design, Machine Learning, Deep Learning, Big Data, Data Structures

Experience

- **CampusMesh** Jan 2026 – Present
Software Engineer
– Built scalable **TypeScript/Node.js microservices** for real-time academic workflows across messaging, calls, notifications, and profile management, backed by **MySQL (Sequelize)** and **DynamoDB**.
– Deployed **serverless infrastructure** on AWS with **Lambda, API Gateway, S3, and CloudFront** across **4 environments**; enforced secure access with **KMS-managed keys, SSM Parameter Store, and least-privilege IAM**.
– Developed **Python-driven LLM retrieval pipelines (RAG)** for CampusIQ, applying **embedding-based semantic search** over structured indexes to improve contextual accuracy by **30%**.
- **Amazon** May 2025 – Aug 2025
Software Development Engineer Intern
Seattle, US
– Provisioned **Infrastructure-as-Code** with **AWS CDK**, deploying API Gateway, Lambda, DynamoDB, S3, and IAM; reduced provisioning time **70%** with **multi-stage, multi-region CI/CD**.
– Built an event-driven platform on **AWS ECS Fargate, SQS, and SNS**, improving incident response latency by **50%** for high-volume operational monitoring workloads.
– Shipped a **React 18** and **TypeScript** frontend with optimized API integration and caching; engineered backend services in **Java** with **Mockito** unit and integration tests.
- **New York University (Novel AI Technologies, Inc.)** Aug 2024 – Present
Software Developer
New York, US
– Engineered parallelized **Python ETL/data pipelines** converting unstructured **NoSQL** into **PostgreSQL** via **GCP Cloud Functions**, with concurrency control and locking; processed **250K+** records for analytics dashboards.
– Built a full-stack platform (**React, TypeScript, Node.js, REST APIs**) deployed via **Docker** on **EC2**; implemented **RBAC**, row-level security, and **WebSocket** real-time updates for multi-tenant clients.
– Deployed cloud-hosted **AI models** on **AWS SageMaker** for health metric analysis and image processing on an NSF-funded mobile application.
- **PricewaterhouseCoopers (PwC)** Aug 2021 – Aug 2024
Senior Software Engineer
Bengaluru, India
– Developed a distributed backend on **Microsoft Azure** with event-driven processing and fault-tolerant retries over **SQL Server** stores.
– Deployed an **NLP-based OCR neural model** on Azure to automate structured data extraction, reducing processing time by **75%**.
– Optimized bulk data ingestion with **parallel processing** on **100K-row** feeds, cutting execution from **5 hours to 1 hour**.

Projects

- **InviLite (AI Monitoring & Data Platform)** | *Python, AWS, Kinesis, Lambda, Glue, SageMaker, Bedrock, FAISS*
– Architected cloud-native **data pipelines** (Kinesis, Lambda, S3, Glue, DynamoDB) ingesting structured and unstructured telemetry into governed **ML workflows**.
– Built a **RAG classification system** (SVM + FAISS + **Bedrock LLM**), cutting incident resolution from **1–2 hours to minutes** and lifting detection accuracy from **56% to 92%**.
- **Hospital Management System (Full-Stack Application)** | *React, TypeScript, Node.js, MongoDB, AWS SageMaker, Hugging Face*
– Built a **React/TypeScript** healthcare platform managing appointments, patient records, and lab reports with **role-based access control** and API-driven workflows.
– Fine-tuned a **Hugging Face transformer model** reaching **88% accuracy** and deployed inference on **AWS SageMaker** for real-time diagnostic support.

Skills

- **Programming Languages:** Python, TypeScript, JavaScript (ES6), Java, SQL, C++
- **Web & Backend:** React, Node.js, Express.js, REST API, GraphQL API, Microservices
- **Databases & Pipelines:** PostgreSQL, MySQL, DynamoDB, MongoDB, ETL/ELT, Data Pipelines
- **Cloud & DevOps:** GCP (BigQuery, Cloud Functions), AWS (CDK, Lambda, ECS, S3, SageMaker, Glue, IAM), Azure, Docker, Kubernetes, CI/CD
- **AI/ML:** PyTorch, Hugging Face, SageMaker, Bedrock, LLM Fine-tuning, RAG (LangChain, FAISS), NLP
- **Tools & Platforms:** Git, Linux, Postman, Jupyter, Visual Studio

Certifications

AWS Cloud Practitioner

Azure Data Fundamentals

Stanford Machine Learning (Coursera)